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I. Multiple Choice Questions

1. In a simple harmonic oscillation, the acceleration against displacement for one complete oscillation will be (model NSEP 2000-01)

a) an ellipse b) a circle

c) a parabola d) a straight line

2. A particle executing SHM crosses points A and B with the same velocity. Having taken 3 s in passing from A to B, it returns to B after another 3 s. The time period is

a) 15 s b) 6 s

c) 12 s d) 9 s

3. The length of a second's pendulum on the surface of the Earth is 0.9 m. The length of the same pendulum on surface of planet X such that the acceleration of the planet X is n times greater than the Earth is

a) $0.9n$ b) $\frac{0.9}{n}m$

c) $0.9n^2m$ d) $\frac{0.9}{n^2}$

4. A simple pendulum is suspended from the roof of a school bus which moves in a horizontal direction with an acceleration a, then the time period is

a) $T\alpha\frac{1}{g^2+a^2}$ b) $T\alpha\frac{1}{\sqrt{g^2+a^2}}$

c) $T\alpha\sqrt{g^2+a^2}$ d) $T\alpha(g^2+a^2)$

5. Two bodies A and B whose masses are in the ratio 1:2 are suspended from two separate massless springs of force constants k_A and k_B respectively. If the two bodies oscillate vertically such that their maximum velocities are in the ratio 1:2, the ratio of the amplitude A to that of B is

a) $\sqrt{\frac{k_B}{2k_A}}$ b) $\sqrt{\frac{k_B}{8k_A}}$

c) $\sqrt{\frac{2k_B}{k_A}}$ d) $\sqrt{\frac{8k_B}{k_A}}$

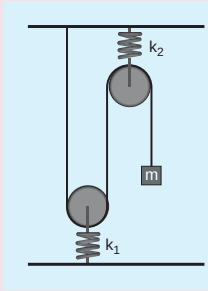
6. A spring is connected to a mass m suspended from it and its time period for vertical oscillation is T. The spring is now cut into two equal halves and the same mass is suspended from one of the halves. The period of vertical oscillation is

a) $T'=\sqrt{2}T$ b) $T'=\frac{T}{\sqrt{2}}$

c) $T'=\sqrt{2T}$ d) $T'=\sqrt{\frac{T}{2}}$

7. The time period for small vertical oscillations of block of mass m when the masses of the pulleys are negligible and spring constant k_1 and k_2 is

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a) $T=4\pi\sqrt{m\left(\frac{1}{k_1}+\frac{1}{k_2}\right)}$

b) $T=2\pi\sqrt{m\left(\frac{1}{k_1}+\frac{1}{k_2}\right)}$

c) $T=4\pi\sqrt{m(k_1+k_2)}$

d) $T=2\pi\sqrt{m(k_1+k_2)}$